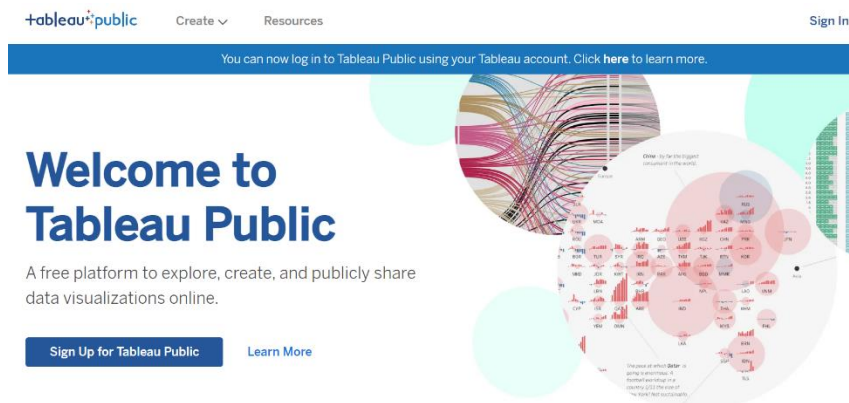


Steps to Create a Data Visualization in Tableau Public

1. Go to Tableau Public Website to start: <https://public.tableau.com/app/discover>
2. Click: “Sign Up for Tableau Public”



3. Create your account. Be sure to pick a password you can remember.

Create your Tableau Account

First Name

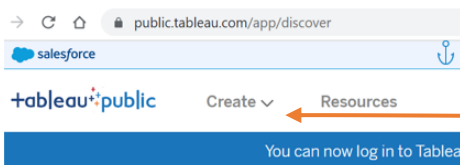
This field is required.

Last Name

4. Fill in all information; you will need to provide a personal email associated with your account, then click “create my account.”

CREATE MY ACCOUNT

5. You will have to verify your email address, so go to your email account and verify before you continue.
6. Once you are signed in, go to “Create” (top left) (website link here: <https://public.tableau.com/app/discover>)

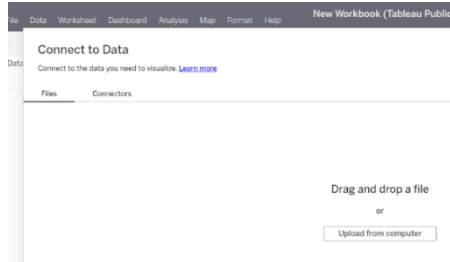


Select “Create” and then menu option “Web Authoring”

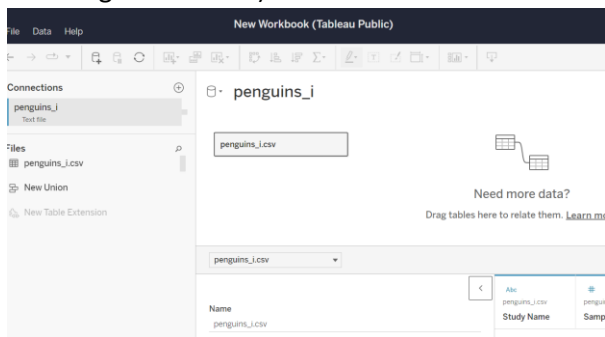
7. Sign in with the login information you provided and then click “Sign In”
8. Select “United States”, agree to terms and conditions, and then click “Complete Sign In”

9. Download the data set and use it to explore. For our activity, we used the penguin data set. You can download it using the instructions here: <https://cran.r-project.org/web/packages/palmerpenguins/readme/README.html>
You can also email me at mais.alraee@montgomerycollege.edu, and I will gladly send it to you.

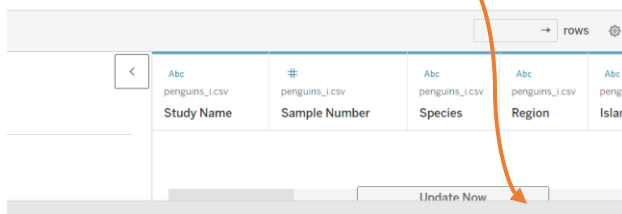
10. Save the **Penguin** dataset to “Downloads” on your computer. Then click “Upload from computer” to load the data:



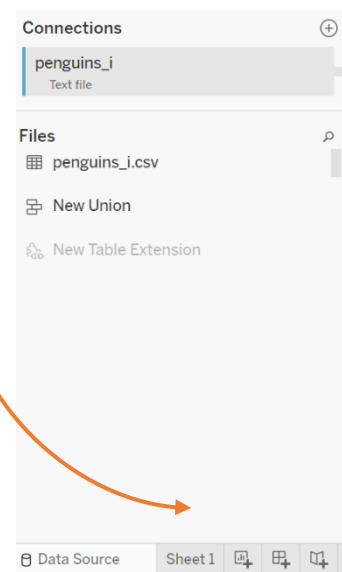
11. Be patient – it takes a few seconds to load. Once it is loaded, your screen will look something like this (I am using the Penguins dataset)



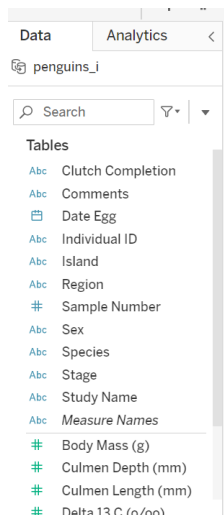
12. On the very bottom, click “update now”



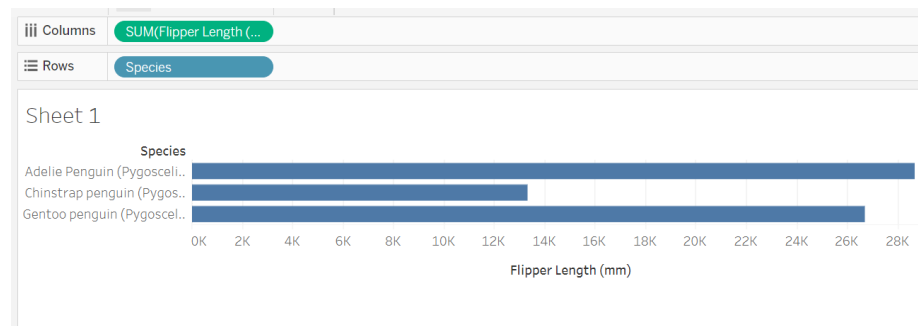
13. Click on the “Sheet 1” tab at the very bottom left:
(or click control z on your keyboard)



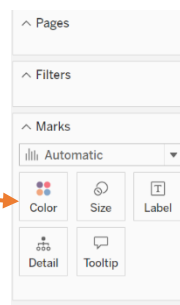
14. On the new page that comes up, you will see all the variables listed on the left:



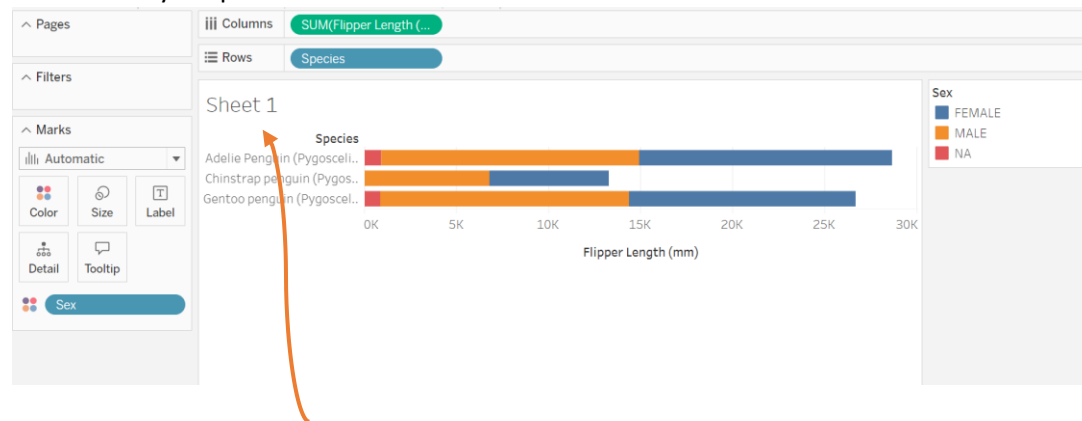
15. Drag “Flipper length” onto the **column shelf** and “Species” onto the **row shelf**



16. On the “Marks”, slide “Sex” into **Color**:

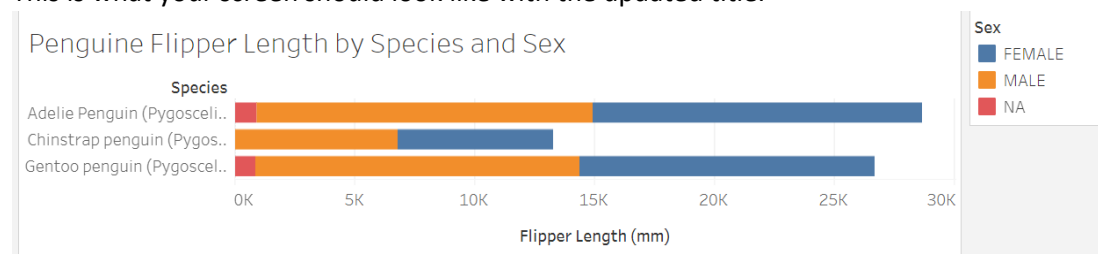


17. This is what your plot should look like now:

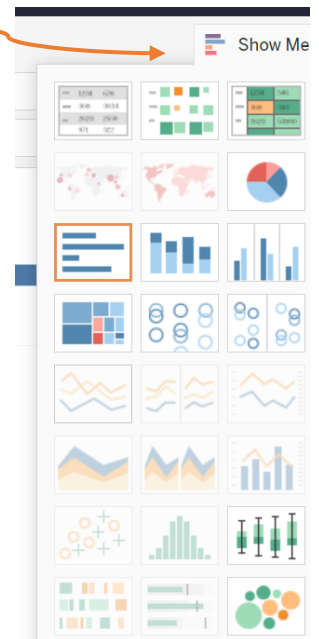


18. Double click on “Sheet 1” above your plot and rename your plot: **Penguin Flipper Length by Species and Sex**

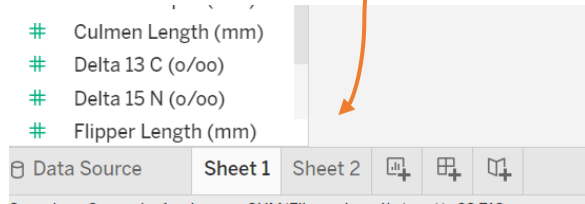
19. This is what your screen should look like with the updated title.



20. Click on “Show Me” on the far top/right to see all data visualization options:

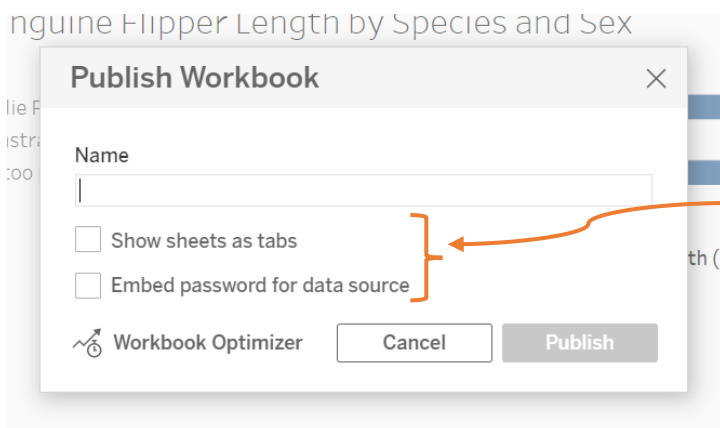


21. On the bottom left, click “**Sheet 2**” to try making a new visualization using different variables and different types of plots from the different “Show Me” options.



22. Slide different types of variables onto the row and column shelves and see what kinds of new visualizations you can create.

23. Once you are done creating your visualizations on one or more “Sheets”, click File → Publish As, it will prompt to create an extract, it might take a while so be patient.



You can leave these two options unchecked.

24. Name your file, anything you choose to name it, and click **Publish**.

Activity Questions:

- 1- What is the average bill length by species and island?
- 2- How does flipper length compare between male and female penguins for each species?
- 3- What is the relationship between body mass and bill depth for each species?
- 4- Which island has the highest average body mass for penguins?
- 5- What is the count of each species on each island

Hypothesis: How might scientists use this kind of data visualization to make decisions about protecting penguin habitats?

Explore: What new variables did you make for your visualization on sheet 2? What did your plot show?